No.	Status	Date
HIS-2006-0412-DS-2-RF-102	Issued	2006-02-24

UT Thickness Data Sheet - 2-RF-102

Scope and Responsibilities

This data sheet provides the ultrasonic thickness measurement results for line 2-RF-102, located within the ATU-100. The scope of work is defined by the 10-year full-unit survey detailed in HIS-2006-0412.

Kevin Doucette, acting as the lead UT technician, is responsible for the execution of the measurements and the recording of the data. All activities were performed in accordance with the technical requirements of the assigned survey.

Technical Basis and Procedures

Thickness measurements were performed under the site NDE procedure QSCP-NDT-001 Rev 0, which governs pre-startup and in-service NDE for the ATU-100. The specific methodology for data acquisition followed HIS-UT-002 Rev 1, as submitted for QSCP acceptance.

The inspection process and instrument calibration were conducted in accordance with the requirements of ASTM E797.

Component Design Data

The subject line, 2-RF-102, is a Reflux line constructed from A312 TP316. The design specifications are an outer diameter of 60.3 mm with a nominal wall thickness of 3.91 mm.

The design limits for this component are 517 kPa(g) and 104 °C. The minimum allowable thickness (t-min) is 2.16 mm.

Integrity Assessment

The retirement thickness for 2-RF-102 is established as the t-min of 2.16 mm. Any measurement recorded below this value shall be processed as an NCR.

The line is categorized as Class 2, which mandates a thickness inspection interval of 10 years and an external visual inspection interval of 5 years.

Thickness readings, measured 2006-02-24

Line 6-SG-101 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-101-01	7.11	3.56	2001-03-02	6.82	6.59	0.053	0.046	>50	2011-02-24	
6-SG-101-02	7.11	3.56	2001-03-02	6.71	6.51	0.054	0.040	>50	2011-02-24	
6-SG-101-03	7.11	3.56	2001-03-02	6.74	6.42	0.066	0.064	43.2	2011-02-24	
6-SG-101-04	7.11	3.56	2001-03-02	6.73	6.28	0.081	0.090	30.1	2011-02-24	
6-SG-101-05	7.11	3.56	2001-03-02	6.68	6.22	0.092	0.092	28.8	2011-02-24	
6-SG-101-06	7.11	3.56	2001-03-02	6.87	6.49	0.058	0.076	38.4	2011-02-24	

Line 6-SG-102 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-102-01	7.11	3.56	2001-03-02	6.72	6.26	0.088	0.092	29.2	2011-02-24	
6-SG-102-02	7.11	3.56	2001-03-02	6.72	6.25	0.086	0.094	28.5	2011-02-24	
6-SG-102-03	7.11	3.56	2001-03-02	6.69	6.26	0.078	0.086	31.3	2011-02-24	
6-SG-102-04	7.11	3.56	2001-03-02	6.64	6.22	0.077	0.084	31.6	2011-02-24	
6-SG-102-05	7.11	3.56	2001-03-02	6.93	6.72	0.041	0.042	>50	2011-02-24	

Line 8-TG-101 | Treated gas | A106 Gr.B | t_nom 8.18 mm | t-min 4.06 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
8-TG-101-01	8.18	4.06	1996-03-08	8.20	7.89	0.031	0.031	>50	2016-02-24	
8-TG-101-02	8.18	4.06	1996-03-08	8.20	7.75	0.045	0.045	>50	2016-02-24	
8-TG-101-03	8.18	4.06	1996-03-08	8.23	7.79	0.044	0.044	>50	2016-02-24	
8-TG-101-04	8.18	4.06	1996-03-08	8.10	7.76	0.034	0.034	>50	2016-02-24	
8-TG-101-05	8.18	4.06	1996-03-08	8.22	7.95	0.027	0.027	>50	2016-02-24	

Line 4-LA-101 | Lean amine supply | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-101-01	6.02	3.18	1996-03-08	5.99	5.51	0.048	0.048	48.4	2016-02-24	
4-LA-101-02	6.02	3.18	1996-03-08	6.01	5.76	0.025	0.025	>50	2016-02-24	
4-LA-101-03	6.02	3.18	1996-03-08	5.96	5.76	0.020	0.020	>50	2016-02-24	
4-LA-101-04	6.02	3.18	1996-03-08	6.07	5.78	0.029	0.029	>50	2016-02-24	
4-LA-101-05	6.02	3.18	1996-03-08	6.01	5.81	0.020	0.020	>50	2016-02-24	
4-LA-101-06	6.02	3.18	1996-03-08	6.06	5.64	0.042	0.042	>50	2016-02-24	
4-LA-101-07	6.02	3.18	1996-03-08	5.99	5.57	0.042	0.042	>50	2016-02-24	

Line 4-LA-102 | Lean amine supply | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-102-01	6.02	3.18	1996-03-08	6.02	5.63	0.039	0.039	>50	2016-02-24	
4-LA-102-02	6.02	3.18	1996-03-08	6.07	5.67	0.040	0.040	>50	2016-02-24	
4-LA-102-03	6.02	3.18	1996-03-08	6.07	5.55	0.052	0.052	45.4	2016-02-24	
4-LA-102-04	6.02	3.18	1996-03-08	5.99	5.34	0.065	0.065	33.1	2016-02-24	
4-LA-102-05	6.02	3.18	1996-03-08	6.00	5.62	0.038	0.038	>50	2016-02-24	
4-LA-102-06	6.02	3.18	1996-03-08	6.00	5.83	0.017	0.017	>50	2016-02-24	
4-LA-102-07	6.02	3.18	1996-03-08	6.00	5.63	0.037	0.037	>50	2016-02-24	

Line 6-LA-103 | Lean amine header | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-LA-103-01	7.11	3.56	1996-03-08	7.02	6.75	0.027	0.027	>50	2016-02-24	
6-LA-103-02	7.11	3.56	1996-03-08	7.19	6.87	0.032	0.032	>50	2016-02-24	
6-LA-103-03	7.11	3.56	1996-03-08	7.08	6.65	0.043	0.043	>50	2016-02-24	
6-LA-103-04	7.11	3.56	1996-03-08	7.17	6.52	0.065	0.065	45.4	2016-02-24	
6-LA-103-05	7.11	3.56	1996-03-08	7.19	6.49	0.070	0.070	41.7	2016-02-24	
6-LA-103-06	7.11	3.56	1996-03-08	7.16	6.42	0.074	0.074	38.5	2016-02-24	
6-LA-103-07	7.11	3.56	1996-03-08	7.18	6.76	0.042	0.042	>50	2016-02-24	
6-LA-103-08	7.11	3.56	1996-03-08	7.04	6.79	0.025	0.025	>50	2016-02-24	

Line 4-RA-101 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-101-01	8.56	3.81	2001-03-02	8.39	8.27	0.025	0.024	>50	2016-02-24	
4-RA-101-02	8.56	3.81	2001-03-02	8.45	8.27	0.027	0.036	>50	2016-02-24	
4-RA-101-03	8.56	3.81	2001-03-02	8.42	8.10	0.051	0.064	>50	2016-02-24	
4-RA-101-04	8.56	3.81	2001-03-02	8.37	8.34	0.029	0.006	>50	2016-02-24	
4-RA-101-05	8.56	3.81	2001-03-02	8.42	8.29	0.031	0.026	>50	2016-02-24	
4-RA-101-07	8.56	3.81	2001-03-02	8.51	8.30	0.030	0.042	>50	2016-02-24	
4-RA-101-08	8.56	3.81	2001-03-02	8.30	8.07	0.061	0.046	>50	2016-02-24	
4-RA-101-09	8.56	3.81	2001-03-02	8.43	8.43	0.014	0.000	>50	2016-02-24	
4-RA-101-10	8.56	3.81	2001-03-02	8.48	8.27	0.040	0.042	>50	2016-02-24	
4-RA-101-11	8.56	3.81	2001-03-02	8.39	8.14	0.054	0.050	>50	2016-02-24	

Line 4-RA-102 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-102-01	8.56	3.81	2001-03-02	8.48	8.41	0.013	0.014	>50	2016-02-24	
4-RA-102-02	8.56	3.81	2001-03-02	8.50	8.15	0.042	0.070	>50	2016-02-24	
4-RA-102-03	8.56	3.81	2001-03-02	8.37	8.27	0.029	0.020	>50	2016-02-24	
4-RA-102-04	8.56	3.81	2001-03-02	8.35	8.24	0.030	0.022	>50	2016-02-24	
4-RA-102-05	8.56	3.81	2001-03-02	8.36	8.19	0.036	0.034	>50	2016-02-24	
4-RA-102-06	8.56	3.81	2001-03-02	8.38	8.15	0.036	0.046	>50	2016-02-24	
4-RA-102-07	8.56	3.81	2001-03-02	8.49	8.29	0.029	0.040	>50	2016-02-24	
4-RA-102-08	8.56	3.81	2001-03-02	8.36	8.05	0.044	0.062	>50	2016-02-24	
4-RA-102-09	8.56	3.81	2001-03-02	8.35	8.18	0.037	0.034	>50	2016-02-24	

Line 6-RA-103 | Rich amine header | A106 Gr.B | t_nom 10.97 mm | t-min 4.57 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-RA-103-01	10.97	4.57	2001-03-02	10.82	10.76	0.024	0.012	>50	2016-02-24	
6-RA-103-02	10.97	4.57	2001-03-02	10.69	10.50	0.059	0.038	>50	2016-02-24	
6-RA-103-03	10.97	4.57	2001-03-02	10.84	10.67	0.032	0.034	>50	2016-02-24	
6-RA-103-04	10.97	4.57	2001-03-02	10.59	10.40	0.058	0.038	>50	2016-02-24	
6-RA-103-05	10.97	4.57	2001-03-02	10.61	10.34	0.061	0.054	>50	2016-02-24	
6-RA-103-06	10.97	4.57	2001-03-02	10.86	10.74	0.027	0.024	>50	2016-02-24	
6-RA-103-07	10.97	4.57	2001-03-02	10.79	10.66	0.033	0.026	>50	2016-02-24	
6-RA-103-08	10.97	4.57	2001-03-02	10.83	10.58	0.039	0.050	>50	2016-02-24	
6-RA-103-09	10.97	4.57	2001-03-02	10.85	10.67	0.034	0.036	>50	2016-02-24	
6-RA-103-10	10.97	4.57	2001-03-02	10.78	10.27	0.075	0.102	>50	2016-02-24	

Line 3-RV-101 | Regenerator overhead | A106 Gr.B | t_nom 7.62 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RV-101-01	7.62	3.56	2001-03-02	7.26	6.98	0.061	0.056	>50	2011-02-24	
3-RV-101-02	7.62	3.56	2001-03-02	7.31	6.93	0.064	0.076	44.2	2011-02-24	
3-RV-101-03	7.62	3.56	2001-03-02	7.15	6.57	0.095	0.116	25.9	2011-02-24	
3-RV-101-04	7.62	3.56	2001-03-02	7.18	6.65	0.094	0.106	29.1	2011-02-24	
3-RV-101-05	7.62	3.56	2001-03-02	7.25	7.01	0.066	0.048	>50	2011-02-24	
3-RV-101-06	7.62	3.56	2001-03-02	7.20	6.74	0.085	0.092	34.4	2011-02-24	

Line 2-RV-102 | Regenerator overhead | A106 Gr.B | t_nom 5.54 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RV-102-01	5.54	2.79	2001-03-02	5.29	4.85	0.068	0.088	23.3	2011-02-24	
2-RV-102-02	5.54	2.79	2001-03-02	5.12	4.84	0.075	0.056	27.2	2011-02-24	
2-RV-102-03	5.54	2.79	2001-03-02	5.24	5.14	0.034	0.020	>50	2011-02-24	
2-RV-102-04	5.54	2.79	2001-03-02	5.33	5.07	0.051	0.052	43.7	2011-02-24	

Line 3-RF-101 | Reflux | A312 TP316 | t_nom 5.49 mm | t-min 2.41 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RF-101-01	5.49	2.41	1996-03-08	5.48	5.48	0.000	0.000	>50	2016-02-24	
3-RF-101-02	5.49	2.41	1996-03-08	5.45	5.37	0.008	0.008	>50	2016-02-24	
3-RF-101-03	5.49	2.41	1996-03-08	5.45	5.45	0.000	0.000	>50	2016-02-24	
3-RF-101-04	5.49	2.41	1996-03-08	5.46	5.33	0.013	0.013	>50	2016-02-24	
3-RF-101-05	5.49	2.41	1996-03-08	5.45	5.40	0.005	0.005	>50	2016-02-24	
3-RF-101-06	5.49	2.41	1996-03-08	5.52	5.40	0.012	0.012	>50	2016-02-24	
3-RF-101-07	5.49	2.41	1996-03-08	5.42	5.35	0.007	0.007	>50	2016-02-24	

Line 2-RF-102 | Reflux | A312 TP316 | t_nom 3.91 mm | t-min 2.16 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RF-102-01	3.91	2.16	1996-03-08	3.92	3.85	0.007	0.007	>50	2016-02-24	
2-RF-102-02	3.91	2.16	1996-03-08	3.88	3.88	0.000	0.000	>50	2016-02-24	
2-RF-102-03	3.91	2.16	1996-03-08	3.93	3.80	0.013	0.013	>50	2016-02-24	
2-RF-102-04	3.91	2.16	1996-03-08	3.87	3.81	0.006	0.006	>50	2016-02-24	
2-RF-102-07	3.91	2.16	1996-03-08	3.96	3.85	0.011	0.011	>50	2016-02-24	

Line 4-LA-104 | L/R exchanger, lean side | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-104-01	6.02	3.18	1996-03-08	6.07	5.75	0.032	0.032	>50	2016-02-24	
4-LA-104-02	6.02	3.18	1996-03-08	6.03	5.56	0.047	0.047	>50	2016-02-24	
4-LA-104-03	6.02	3.18	1996-03-08	5.96	5.74	0.022	0.022	>50	2016-02-24	
4-LA-104-04	6.02	3.18	1996-03-08	6.06	5.86	0.020	0.020	>50	2016-02-24	
4-LA-104-05	6.02	3.18	1996-03-08	6.04	5.51	0.053	0.053	43.8	2016-02-24	
4-LA-104-06	6.02	3.18	1996-03-08	6.10	5.83	0.027	0.027	>50	2016-02-24	
4-LA-104-07	6.02	3.18	1996-03-08	6.01	5.62	0.039	0.039	>50	2016-02-24	
4-LA-104-08	6.02	3.18	1996-03-08	5.94	5.50	0.044	0.044	>50	2016-02-24	

Line 4-RA-104 | L/R exchanger, rich side | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-104-01	8.56	3.81	2001-03-02	8.40	8.16	0.035	0.048	>50	2016-02-24	
4-RA-104-02	8.56	3.81	2001-03-02	8.41	8.23	0.032	0.036	>50	2016-02-24	
4-RA-104-03	8.56	3.81	2001-03-02	8.46	8.30	0.024	0.032	>50	2016-02-24	
4-RA-104-04	8.56	3.81	2001-03-02	8.35	8.21	0.038	0.028	>50	2016-02-24	
4-RA-104-05	8.56	3.81	2001-03-02	8.29	8.14	0.046	0.030	>50	2016-02-24	
4-RA-104-06	8.56	3.81	2001-03-02	8.39	8.09	0.050	0.060	>50	2016-02-24	
4-RA-104-07	8.56	3.81	2001-03-02	8.29	7.97	0.062	0.064	>50	2016-02-24	

Line 2-ST-101 | Reboiler steam | A106 Gr.B | t_nom 3.91 mm | t-min 2.29 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-ST-101-01	3.91	2.29	1996-03-08	3.93	3.64	0.029	0.029	46.4	2016-02-24	
2-ST-101-02	3.91	2.29	1996-03-08	3.90	3.67	0.023	0.023	>50	2016-02-24	
2-ST-101-03	3.91	2.29	1996-03-08	3.94	3.82	0.012	0.012	>50	2016-02-24	
2-ST-101-04	3.91	2.29	1996-03-08	3.82	3.77	0.005	0.005	>50	2016-02-24	
2-ST-101-05	3.91	2.29	1996-03-08	3.91	3.72	0.019	0.019	>50	2016-02-24	

Line 3-CW-101 | Amine cooler CW | A106 Gr.B | t_nom 5.49 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-CW-101-01	5.49	2.79	1996-03-08	5.42	5.13	0.029	0.029	>50	2016-02-24	
3-CW-101-02	5.49	2.79	1996-03-08	5.41	5.20	0.021	0.021	>50	2016-02-24	
3-CW-101-03	5.49	2.79	1996-03-08	5.53	5.08	0.045	0.045	>50	2016-02-24	
3-CW-101-04	5.49	2.79	1996-03-08	5.46	5.38	0.008	0.008	>50	2016-02-24	
3-CW-101-05	5.49	2.79	1996-03-08	5.48	5.33	0.015	0.015	>50	2016-02-24	

Line 3-CW-102 | Amine cooler CW | A106 Gr.B | t_nom 5.49 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-CW-102-01	5.49	2.79	1996-03-08	5.50	5.20	0.030	0.030	>50	2016-02-24	
3-CW-102-02	5.49	2.79	1996-03-08	5.51	5.02	0.049	0.049	45.4	2016-02-24	
3-CW-102-03	5.49	2.79	1996-03-08	5.53	5.07	0.046	0.046	49.4	2016-02-24	
3-CW-102-04	5.49	2.79	1996-03-08	5.51	5.03	0.048	0.048	46.5	2016-02-24	

Sign-off	Name	Function	Date	Signature
Prepared	Kevin Doucette	lead UT technician, SNT-TC-1A UT Level II	2006-02-24	_____